

Week ①

linear equation

linear algebra 1 & 2

system of sentences

equation $\Rightarrow$ lines & Planes

Singular & non singular

linear dependence & independence

determinant

① linear algebra

linear regression: collect data (i/p and o/p) and you need to find relation between them

ex $y = mx + b$

weight

bias

For more inputs

ex

Wind speed

Temp

Pressure

humidity

$w_1 x_1$

$w_2 x_2$

$w_3 x_3$

$w_4 x_4$

$w_1 x_1^2$

$w_2 x_2^2$

$w_3 x_3^2$

$w_4 x_4^2$

$\vdots$

$\vdots$

$\vdots$

$\vdots$

$w_1 x_1^m$

$w_2 x_2^m$

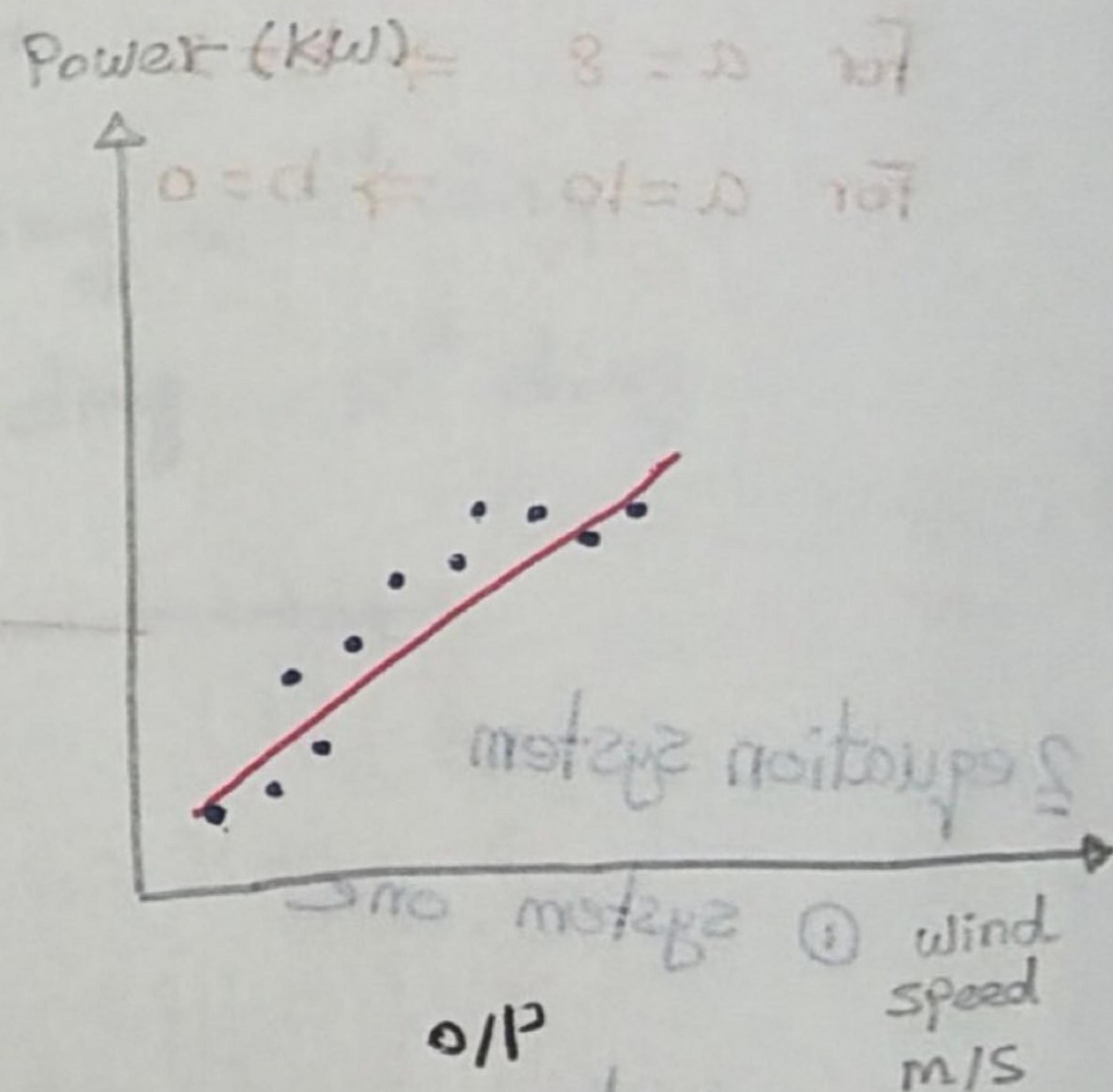
$w_3 x_3^m$

$w_4 x_4^m$

$w \cdot x + b = \hat{y}$

$[y^1 \ y^2 \ \dots \ y^m]$

vector



Target

y

y^2

y^m

linear equation

- is an equation where var appears only to Pwr 1 and var not multiplied by each other.

ex $y + 2x = 3$
 $3x + 2y = 7$

non linear equation:

$y = x^2 + 21$

$xy = 5$

$y = \frac{1}{x}$

$y = \sqrt{x}$

and For graph shape

- ① Curved
- ② Parabolic
- ③ exp
- ④ Circular

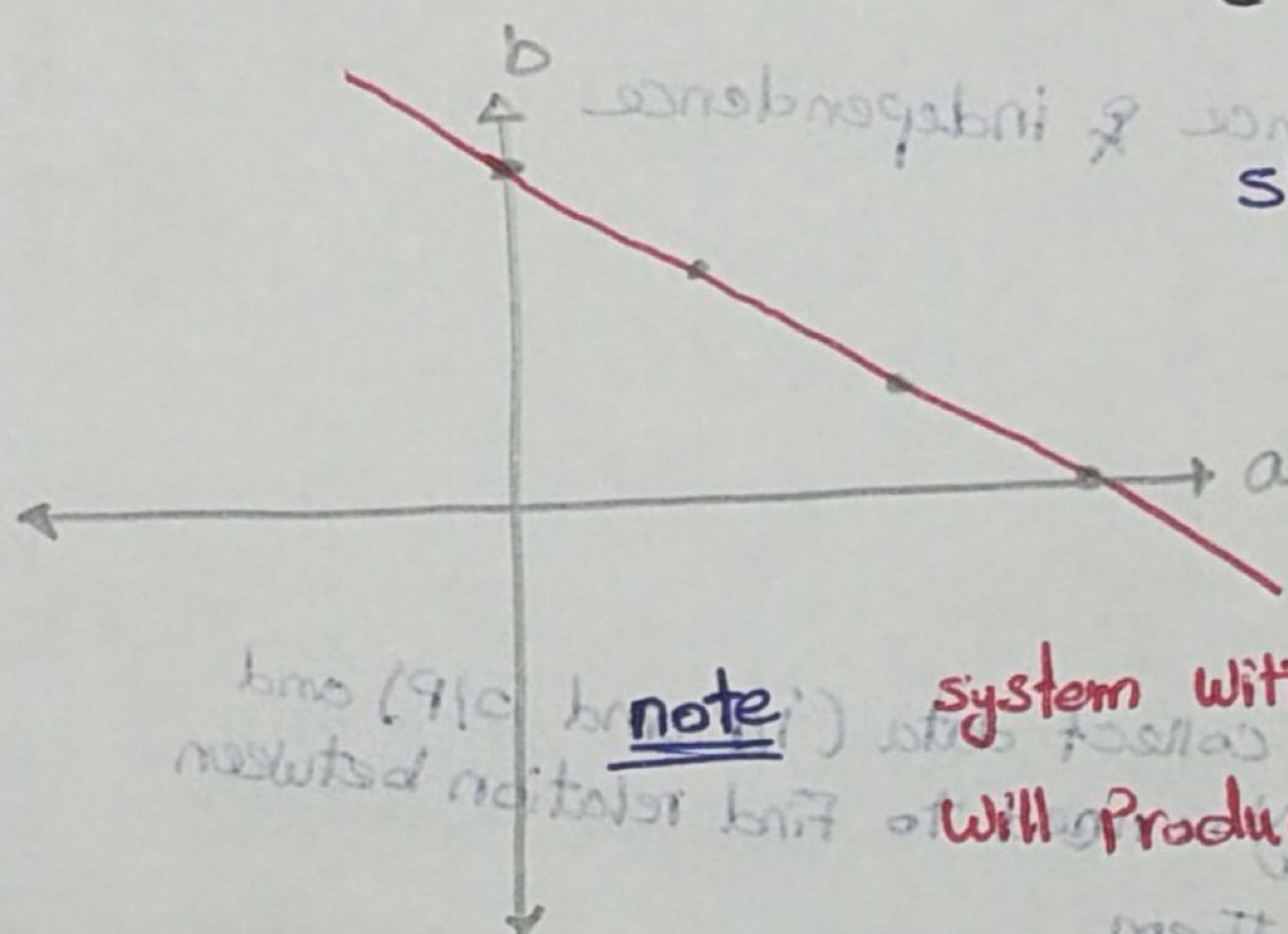
ex $a + b = 10$

For $a = 0 \Rightarrow b = 10$

For $a = 4 \Rightarrow b = 6$

For $a = 8 \Rightarrow b = 2$

For $a = 10 \Rightarrow b = 0$



slope = $\frac{y_2 - y_1}{x_2 - x_1}$
 slope = $\frac{2 - 10}{8 - 0} = -1$

note system with 2 equations will produce 2 lines

2 equation system

① system one

$a + b = 0$

$a + 2b = 0$

② system 2

$a + b = 0$

$2a + 2b = 0$

• system 2 is linear dependant because they carry same info

how to determine is system is linear dependant or non independant?

it will be relation between Rows

ex

$a + b + c = 0$

$2a + 2b + 2c = 0$

$3a + 3b + 3c = 0$

$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$

$\therefore \text{row 1} + \text{row 2} = \text{row 3}$

$\therefore$ it is linear dependant.

note Sometimes you should do mathematical operation to check

if it linear dependant or non dependant ex $3\text{row}_2 + 2\text{row}_3 = \text{row}_1$

* Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det = AD - CB$$

if $\det = 0 \Rightarrow$ matrix is singular

if $\det \neq 0 \Rightarrow$ matrix is non singular

For 3×3 matrix

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & k \end{bmatrix}$$

$$1^{\text{st}} \text{ diag} = aek$$

$$2^{\text{nd}} \text{ diag} = bfg$$

$$3^{\text{rd}} \text{ diag} = cdh$$

$$4^{\text{th}} \text{ diag} = ceg$$

$$5^{\text{th}} \text{ diag} = afh$$

$$6^{\text{th}} \text{ diag} = bdk$$

$$\det = 1^{\text{st}} \text{ diag} + 2^{\text{nd}} \text{ diag} + 3^{\text{rd}} \text{ diag} - 4^{\text{th}} \text{ diag} - 5^{\text{th}} \text{ diag} - 6^{\text{th}} \text{ diag}$$

Numpy

numpy is main package for scientific computing in python.

You can:

① Create array

② slicing

③ indexing

④ reshaping

⑤ stacking

$\Rightarrow$ Same data type unlike list.
numpy is faster than list

example of numpy object:

`np.array([10, 12])` // 1-D array with 2 value

`np.arange(3)` // Create 3 values with 0

`np.arange(1, 20, 3)` // Create array that start from 1 to 20 increment by 3

`[1 4 7 10 13 16 19]`

`np.linspace(0, 100, 5)` // Create 5 numbers evenly spaced value between 0 to 100.

default data type is Float64 to be int $\Rightarrow$ `np.linspace(0, 100, 5, dtype=int)`

more func on numpy

① np.ones(3) $\Rightarrow$ Create 3 elements = 1

② np.zeros(3) $\Rightarrow$ " 3 " = 0

③ np.empty(3) $\Rightarrow$ " 3 " with no init

④ np.random.rand(3) $\Rightarrow$ 3 elements with random values between 0 & 1

⑤ Determinant

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

⑥ multi dimensional array

① Create 2 dimensional array $\Rightarrow$ np.array([1, 2, 3], (5, 7, 8))

$$\begin{bmatrix} 1, 2, 3 \\ 5, 7, 8 \end{bmatrix}$$

② reshape 1D array to 2D array

one_dim_arr = np.array([1, 2, 3, 4, 5, 6])

mul_dim_arr = np.reshape(one_dim_arr, (2, 3))

$$\begin{bmatrix} 1, 2, 3 \\ 4, 5, 6 \end{bmatrix}$$

For get array dimension

$\Rightarrow$ mul_dim_arr.ndim // 2

For get shape of arr

$\Rightarrow$ mul_dim_arr.shape // (2, 3)

For get size of arr

$\Rightarrow$ mul_dim_arr.size // 6

③ math operation

arr-1 = np.array([2, 4, 6])

arr-2 = np.array([1, 3, 5])

① add addition = arr-1 + arr-2

② sub subtract = arr-1 - arr-2

③ mul multiplication = arr-1 * arr-2

④ multiply vector with scalar (broadcast)

vector = np.array([1, 2])

vector * 2.5

⑤ indexing & slicing

arr = np.array([2, 4, 6])

Print(a[2]) // 6

arr-2D = np.array([1, 2, 3],

[4, 5, 6],

[7, 8, 9])

Print(a[2][1]) // 8

For sliced: arr = np.array([1, 2, 3, 4, 5])

sliced_arr = arr[:3] // [1, 2, 3]

sliced_arr = arr[1:4] // [2, 3, 4]

sliced_arr = arr[:, :2] // [1, 3, 5]

⑥ stacking

a1 = np.array([1, 1], [2, 2])

a2 = np.array([3, 3], [4, 4])

vert_stack = np.vstack(a1, a2)

// [[1, 1]

[2, 2]

[3, 3]

[4, 4]]

horz_stack = np.hstack(a1, a2)

// [[1 1 3 3] [2 2 4 4]]

Lab 2 week 1 linear system as matrices

① Packages usage

```
import numpy as np
import matplotlib.pyplot as plt
from utils import plot_lines
```

system of linear equation
$$\begin{cases} -x_1 + 3x_2 = 7 \\ 3x_1 + 2x_2 = 1 \end{cases}$$

system of linear equation as matrices
$$\begin{bmatrix} -1 & 3 & 7 \\ 3 & 2 & 1 \end{bmatrix}$$

so matrix A
$$\begin{bmatrix} -1 & 3 \\ 3 & 2 \end{bmatrix}$$

matrix B
$$\begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

using np:

```
A = np.array([-1, 3], [3, 2])
```

```
b = np.array([7, 1])
```

① Solving equation

np linear algebra Provide quick and reliable way to solve equation using `np.linalg.solve(A, b)`

```
x = np.linalg.solve(A, b)
```

```
Print(x) // [-1 2]
           ↙   ↘
          x1  x2
```

note you can calc det and plot_lines

for ① system with Finite solution as shown in previous example

② system with no solution
ex
$$\begin{cases} -x_1 + 3x_2 = 7 \\ 3x_1 - 9x_2 = 1 \end{cases}$$

③ system with infinite solution
ex
$$\begin{cases} -x_1 + 3x_2 = 7 \\ 3x_1 - 9x_2 = -21 \end{cases}$$

② Evaluate determinant

```
d = np.linalg.det(A)
```

```
Print(d) // -11
```

③ Visualising 2x2 matrix as Plotlines

```
b = b.reshape((2, 1))
```

```
A_system = np.hstack(A, b)
```

```
[[[-1 3 7]
 [3 2 1]]
```

```
Plot_lines(A_system)
```

here, 2 lines intersect on Point $(-1, 2)$

↙ ↘
x₁ x₂